

Class 23

Money, Banking, and Crises

Part IV — Governing Capitalism's Consequences

The American Economy — draft course materials (proposal under discussion)

The point of today's class

Modern capitalism needs banks because they turn savings into credit and liquidity — but banks are fragile for exactly the same reason: they borrow short and lend long. Central banks and deposit insurance exist to stop private fear from becoming economy-wide collapse.

Why it matters

- ▶ Pull out a dollar. Why will a stranger hand you real goods for a piece of paper? Not gold — **trust**. Money is trust made transferable.
 - ▶ Money does three jobs — medium of exchange, unit of account, store of value — and all three run on the belief that others will accept it.
- ▶ Your bank balance is a promise too, and promises can be doubted.
- ▶ American history is a drumbeat of doubted promises: panics in 1837, 1857, 1873, 1893, 1907, 1930–33, 2008, 2023.
- ▶ Today: why the same institution that creates credit also creates the possibility of panic — and what the Fed and the FDIC are for.

Today

1. Banks: useful and fragile for the same reason
2. The bank-run game: can you kill a healthy bank?
3. America before the Fed: a panic every twenty years
4. Bagehot and the lender of last resort
5. The Great Depression as monetary failure — and the FDIC
6. 2008: the run moved to the shadows
7. The modern Fed: inflation, unemployment, interest rates

Banks are not vaults

- ▶ The usual mental model — the bank keeps your cash in a vault — is wrong. A bank keeps a sliver as reserves and lends the rest.

Liabilities (what the bank owes)	Assets (what the bank owns)
Checking and savings deposits	Mortgages
Short-term borrowing	Business loans
Bank equity	Treasury securities
	Reserves and cash

- ▶ That is exactly what makes banks **useful**: payments, credit for homes and businesses, liquidity for depositors, screening and monitoring of borrowers.

Borrow short, lend long: maturity transformation

- ▶ Depositors want money *now*; borrowers want years to repay. Banks bridge the two — that is the service, and the danger.
- ▶ Example: \$100 in deposits, \$10 in reserves, \$90 lent out as mortgages and business loans. A few withdrawals: fine. Everyone at once: mortgages cannot be turned back into cash today.

Illiquidity	Insolvency
Cannot get cash fast enough Sound in the long run	Assets worth less than liabilities Broke in any run

- ▶ Panic is dangerous because it converts one into the other: fire-sale prices turn illiquid banks insolvent — and spread the fear to the next bank.

Game: can you kill a healthy bank?

- ▶ You are all depositors with \$100 at First National. The bank is **solvent** — its loans are good — but only 20% of deposits are on hand as cash today.
- ▶ Each round, write down one choice: **STAY** or **WITHDRAW**.
 - ▶ If few withdraw: withdrawers get \$100; stayers keep earning interest.
 - ▶ If too many withdraw: the bank fire-sells its loans — the first in line get \$100, everyone after splits what is left.
- ▶ Round 1: no news.
- ▶ Round 2: a rumor spreads that the bank “made some bad loans.” (The rumor is false, and you are told it is unverified.)
- ▶ Round 3: you see how many people withdrew last round.

What the game teaches

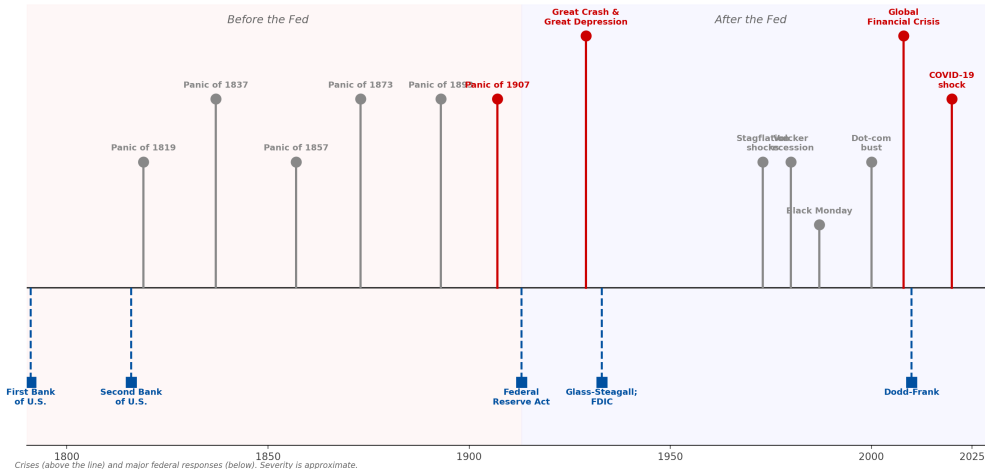
- ▶ Nothing about the bank changed between rounds — only beliefs about *other depositors*.
- ▶ Diamond–Dybvig in plain English: two self-fulfilling equilibria.

If everyone stays calm	If everyone runs
Bank keeps lending	Bank fire-sells assets
Depositors get paid	First in line get paid
Credit continues	Credit collapses

- ▶ If you believe others will run, running is *individually rational* — fear creates the thing feared.
- ▶ A bank run is a coordination failure: a healthy bank can die from fear alone.

Crises and the case for a lender of last resort

U.S. banking and financial crises, and the institutional response



Source: Federal Reserve Bank of New York; Reinhart and Rogoff (2009). Panics of 1837–2023.

America before the Fed: a panic roughly every twenty years

- ▶ Fragmented banks, seasonal cash demand, no central bank after the Second Bank lost its charter — and no official lender of last resort.

Episode	Lesson
Panic of 1837	Credit booms can end in long depressions
Panic of 1857	Shocks travel through trade and railroads
Panic of 1893	Banking, railroads, gold, and confidence collide
Panic of 1907	J.P. Morgan personally organizes the rescue

- ▶ 1907 posed the question: should a modern economy depend on one private banker to stop a national panic?
- ▶ The answer: the **Federal Reserve Act of 1913** — an elastic currency and a public lender of last resort.

Bagehot's rule: the lender of last resort

- ▶ The classic principle for a central bank facing panic:

Rule	Why
Lend freely	Stop the panic; make liquidity available
Against good collateral	Support the illiquid, not the insolvent
At a penalty rate	Do not subsidize recklessness
During panic	Act when private credit freezes

- ▶ The tradeoff is immediate: rescues stop fire sales but create **moral hazard** — expected rescue invites more risk-taking.
- ▶ The Fed is a fire department — but if everyone knows the fire department will always come, some people build riskier houses.

The Great Depression: the Fed fails its first great test

- ▶ The Depression was not just “the stock market crashed” — it was a collapse of money, banking, credit, and confidence.
- ▶ The doom loop: banking panics → banks fail or hoard → depositors hoard cash → money supply collapses → deflation raises real debt burdens → more failures and unemployment.
- ▶ From fall 1930 to winter 1933 the money supply fell by nearly 30%.
- ▶ Friedman–Schwartz and Bernanke: the Fed stood by — the lender of last resort did not lend.
- ▶ The institutional response, 1933: the **FDIC**. Insured depositors have no reason to run — deposit insurance is anti-panic technology. The cost: less monitoring, more risk-taking, and a bigger government backstop — so regulation and supervision follow.

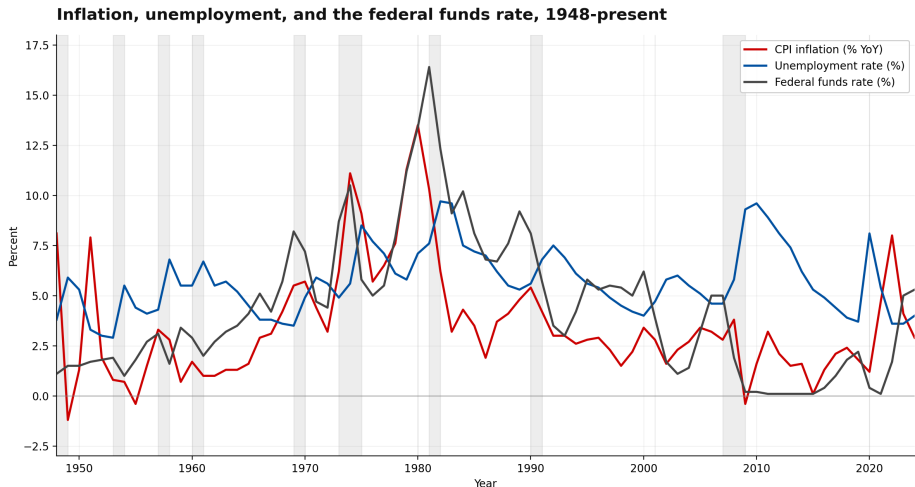
2008: the run moved to the shadows

- ▶ Deposit insurance ended old-fashioned runs — so bank-like finance grew up outside insured banking.

Traditional bank	Shadow bank
Deposits	Repo and wholesale funding
Loans	Mortgage-backed securities
Deposit insurance	Implicit or fragile guarantees
Fed discount window	Facilities improvised mid-crisis

- ▶ 2008 in one line: housing losses → nobody knows who holds the bad assets → repo lenders and money funds pull back → fire sales, credit crunch, recession.
- ▶ Same Diamond–Dybvig logic in new clothes — plus a new moral-hazard problem: *too big to fail*.

Inflation, unemployment, and the federal funds rate



Source: Federal Reserve Economic Data: CPIAUCSL, UNRATE, FEDFUNDS.

Discussion

Four institutions are in trouble: a small solvent bank facing a panic; a large bank insolvent after bad bets; a money-market fund whose run threatens the payment system; a crypto exchange whose customers lost money. Which, if any, should the Fed rescue — and what rule would you write so that rescue today does not invite recklessness tomorrow?

Read before next class

- ▶ Bernanke, “Money, Gold, and the Great Depression” — an accessible account of the Fed’s failure.
- ▶ Federal Reserve History, “The Panic of 1907” — the crisis that created the Fed (short read).